

B. Tech Degree IV Semester (Supplementary) Examination January 2011

ME 404 APPLIED THERMODYNAMICS (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A(Answer ALL questions)

(8 x 5 = 40)

- I. (a) Explain briefly the First Law of Thermodynamics for a cyclic process.
 (b) What are Helmholtz and Gibbs's functions?
 (c) Explain with a neat diagram p-V-T surface.
 (d) State the advantages of Regenerative cycle over simple Rankine cycle.
 (e) Derive a relation between volumetric and gravimetric analysis.
 (f) Distinguish between reciprocating and rotary air compressors.
 (g) Enumerate the characteristics of an ideal fuel.
 (h) Estimate the higher and lower calorific values of a fuel having the following composition by mass: C = 88%, H₂ = 10%, O₂ = 3%, S = 2%, N₂ = 20% and the rest is incombustible matter.

PART B

(4 x 15 = 60)

- II. (a) Prove that the efficiency of an engine working on a reversible cycle depends only on the temperature of source and sink and is independent of the working fluid. (7)
 (b) (i) A reversible heat pump is used to maintain a temperature of 0°C in a refrigerator when it rejects the heat to the surroundings at 25°C. If the heat removal rate from the refrigerator is 1440KJ/Min, determine the C.O.P of the machine and work input required.
 (ii) If the required input to run the pump is developed by a reversible engine which receives heat at 380°C and rejects heat to atmosphere, then determine the overall C.O.P of the system. (8)

OR

- III. (a) Explain the concept of Clausius inequality. (7)
 (b) A reversible heat engine receives heat from two thermal reservoirs maintained at constant temperatures of 750K and 500K. The engine develops 100kw and rejects 3600kJ/min. of heat to a heat sink at 250k. Determine thermal efficiency of the engine and heat supplied by each thermal reservoir. (8)

- IV. (a) Describe the different processes of Rankine cycle. Represent it on P-V and T-S diagram. Also derive the expression for its efficiency. (7)
 (b) A simple Rankine cycle works between pressure 28 bar and 0.06 bar, the initial condition of steam being dry saturated. Calculate (i) cycle efficiency (ii) work ratio (iii) specific steam consumption. (8)

OR

- V. (a) Explain with a neat diagram the working of a reheat cycle. Show the ideal reheating process on T-S and h-S chart. (7)
 (b) A steam power plant operates on a theoretical reheat cycle. Steam at boiler at 150 bar, 550°C expands through the high pressure turbine. It is reheated at a constant pressure of 40 bar to 550°C and expands through the low pressure turbine to condenser at 0.1 bar. Draw T-S and h-S diagram. Find: (i) Quality of steam at turbine exhaust (ii) Cycle efficiency (iii) Steam rate in kg/kwh. (8)

(P.T.O)

- VI. (a) Prove $M = \frac{1}{\sum \frac{m_f}{M_i}}$, where $M \rightarrow$ Molecular weight of the mixture,
 m_f – mass fraction of the constituent, M_i – molecular weight of the constituent. (7)

- (b) A vessel of 0.35 m^3 capacity contains 0.4 kg of carbon monoxide (Molecular weight = 28) and 1 kg of air at 20°C .

- Calculate (i) The partial pressure of each constituent
(ii) The total pressure in the vessel.

The gravimetric analysis of air is to be taken as 23.3% oxygen (molecular weight = 32) and 76.7% Nitrogen (molecular weight = 28). (8)

OR

- VII. (a) Prove that the work done/kg of air in a reciprocating compressor is give by

$$W = RT_1 \frac{n}{n-1} \left[\left(r_p \right)^{\frac{n-1}{n}} - 1 \right]$$

where r_p = Pressure ratio. (7)

- (b) Following data relate to a performance test of a single – acting $14 \text{ cm} \times 10 \text{ cm}$ reciprocating compressor.

Suction pressure	=	1 bar
Suction temperature	=	20°C
Discharge pressure	=	6 bar
Discharge temperature	=	180°C
Speed of compressor	=	1200 rpm
Shaft power	=	6.25 kw
Mass of air delivered	=	1.7 kg/min.

Calculate the following:

- (i) The actual volumetric efficiency
(ii) The indicated power
(iii) The Isothermal efficiency
(iv) The mechanical efficiency
(v) The overall isothermal efficiency (8)

- VIII. (a) Explain briefly (i) Enthalpy of formation (ii) Adiabatic flame temperature. (7)
(b) Explain with a neat diagram the working of Bomb calorimeter. (8)

OR

- IX. (a) Define: (i) Theoretical air and excess air (ii) Stoichiometric air fuel (A/F) ratio (iii) Calorific or heating values of fuels. (6)
(b) The following is the ultimate analysis of a sample of petrol by weight:
Carbon = 85 percent, Hydrogen = 15 percent. Calculate the ratio of air to petrol consumption of weight if the volumetric analysis of the dry exhaust gas is:
 $\text{CO}_2 = 11.5$ percent, $\text{O}_2 =$ percent, $\text{N}_2 = 86$ percent. Also find percentage excess air. (9)